



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT (24CI3201)

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TITLE OF THE PROJECT	AI-Powered Dataset Quality Analysis for Machine Learning Pipelines	
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Abstract

Problem

Poor-quality datasets can negatively affect the accuracy and reliability of Machine Learning models. Identifying missing values, duplicates, outliers, and inconsistent data manually is time-consuming.

Proposed Solution

The project provides an **AI-powered system** that automatically analyzes uploaded datasets and identifies common data-quality issues before they enter a Machine Learning pipeline.

Key Features

- Detects missing and duplicate values
- Identifies incorrect data types and outliers
- Analyzes class imbalance
- Generates an overall **dataset-quality score**
- Provides visual data-quality reports
- Uses AI to explain detected issues

- Suggests suitable data-preprocessing improvements

Technology

The system uses **Python, Pandas, NumPy, Scikit-learn, Flask/FastAPI, and an AI API** for analysis and recommendations.

Outcome

The system reduces manual dataset inspection and helps users **identify and resolve data-quality problems before model training**, resulting in more reliable Machine Learning pipelines.

Signature of the Guide

HOD